LOAD BALANCING

What is Load Balancer

1. a load balancer acts as the traffic cop sitting Infront of our server and routing client request across all server capable of fulfilling those requests.
2. A Load Balancer is a service that distributes incoming network traffic across multiple servers so that no single server gets overloaded.

3. **it generally provides**
    a. High availability --> If one server fails, traffic goes to others
    b. fault tolerance ----> System keeps running even if some instances fail
    c. Scalability ----------> Easily add/remove servers
    d. better Performance ---> Traffic is evenly distributed

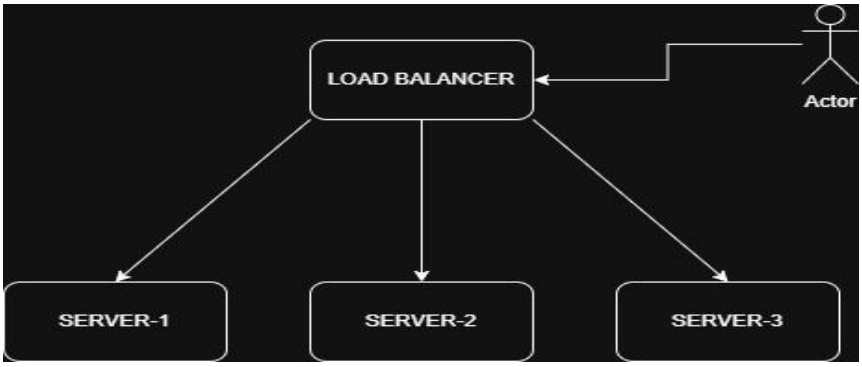
4. **it has three types**
    a. application load balancer (HTTP/HTTPS)
    b. network load balancer (TCP/UDP)
    c. getaway load balancer

5. **main components of the load balancer are**
    1.listener
    listener can listen the request and check where it can go.it can check whether the request is for port no 80, port no 3389 or port no 443 and redirect this request to the target group.
    2.target group
    it is a cluster, a group of instances where we can add no of instances so that they can attached to the load balancer with help of target group, load balancer can only distribute traffic to the instances that are in target group.
    3.target
    target is nothing but a individual instance in target group.

6. and generally, it is recommended that you can select maximum availability zone to places your instances in load balancer for high availability and fault tolerance.

7. **Health check in load balancer**
    a. respond time out is (5 sec) -> (range 2-60 sec)
    b. healthy period time (default 30 sec) -> (range 5-300sec)
    c. unhealthy threshold (default 2) -> (range 2-10)
    If server fails 2 times → marked unhealthy
    d. healthy threshold (default 5) -> (range 2-10)
    Server must pass 5 checks to become healthy again

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APPLICATION LOAD BALANCER

application load balancer distributes incoming application traffic across multiple targets such as ec2 instances in multiple availability zone.

1. an application load balancer makes routing decision at application layer HTTP/HTTPS.
2. it works on the OSI model of layer 7 i.e. called application layer (HTTP/HTTPS)
3. ALB uses target groups to route traffic to EC2 instances, containers, and IP addresses.
4. It supports cross-zone load balancing by default.
5. It uses a DNS name instead of static IP (no Elastic IP).
6. ALB uses Elastic IP addresses (EIPs) for its frontend.
7. It supports SSL/TLS termination (offloading).
8. ALB performs health checks at the application layer to ensure that traffic is sent only to healthy instances.
9. ALB supports HTTP, HTTPS, and WebSocket protocols.

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NETWORK LOAD BALANCER

network load balancer makes routing decision at transport layer i.e the osi model of layer 4 and it has ability to handle volatile workloads and scale to millions of requests per second.

1. NLB distributes traffic at the transport layer (Layer 4) using TCP, UDP, and TLS.
2. It is designed to handle millions of requests per second with very low latency.
3. it works on the OSI model of layer 4 i.e (transport layer)
4. while NLB uses target groups to route traffic to IP addresses and instances.
5. NLB supports cross-zone load balancing (optional, can be enabled/disabled).
6. It provides static IP addresses (one per Availability Zone) and supports Elastic IP.
7. NLB supports TLS termination (but not advanced HTTP routing like ALB).

8. It performs health checks at Layer 4 (TCP) and optionally HTTP/HTTPS.
9. It supports TCP, UDP, and TLS protocols.
10. It is best suited for high-performance, real-time, and volatile workloads.

GETAWAY LOAD BALANCER

A Gateway Load Balancer (GWLB) is a service provided by Amazon Web Services (AWS) that helps you easily deploy, scale, and manage your third-party virtual appliances like firewalls, intrusion detection systems, etc.

1. Works at OSI Layer 3 and handles traffic using IP packets.
2. Used to deploy and manage security appliances like firewalls and IDS/IPS.
3. Provides transparent traffic inspection without changing source or destination IP.
4. Uses GENEVE protocol (port 6081) for traffic encapsulation.
5. Automatically scales and ensures high availability.
6. Uses Gateway Load Balancer Endpoints (GWLBE) for secure traffic routing.
7. GWLB is used to inspect and secure traffic.

All traffic → Firewall → Then server

Feature	**ALB**	**NLB**	**GWLB**
Layer	7	4	3
Use	Website traffic	High-speed traffic	Security
Protocol	HTTP/HTTPS	TCP/UDP	IP
Special Feature	Smart routing	Very fast	Traffic inspection

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LOAD BALANCER ALGORITHMS

1. ROUND ROBIN

Load balancer sends requests to each server one by one in sequence.

- Request 1 → Server1
- Request 2 → Server2
- Request 3 → Server3
- Request 4 → Server1

Suppose you have 3 web servers:

- Web1
- Web2
- Web3

If 6 users visit the website:

- User1→Web1
- User2→Web2
- User3→Web3
- User4→Web1
- User5→Web2
- User6→Web3

1. Traffic is distributed equally.
2. Every server gets turns one by one.
3. Best when all servers have same capacity.
4. Very simple and commonly used.

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2. WEIGHTED ROUND ROBIN

1. load balancer decides which server having the maximum capacity
2. Load balancer gives more requests to stronger servers and fewer requests to weaker servers.

Suppose
Server1 = 8GB RAM → Weight 3
Server2 = 4GB RAM → Weight 2
Server3 = 2GB RAM → Weight 1

Traffic distributions

1. Request1 → Server1
2. Request2 → Server1

3. Request3 → Server1
4. Request4 → Server2
5. Request5 → Server2
6. Request6 → Server3

- Powerful servers handle more traffic.
- Weak servers handle less traffic.
- Used when servers have different capacities.
- Improves performance and resource usage.

IP HASH

Load balancer sends the same user to the same server based on the user's IP address.

- User IP = 192.168.1.10 → Always Server1
- User IP = 192.168.1.20 → Always Server2

If a user logs in to a website, their session stays on the same server.

1. Rahul → Server1
2. Rahul again → Server1

3. Aman → Server2
4. Aman again → Server2

- Same user always connects to same server.
- Useful for login sessions.
- Prevents session loss.
- Based on client IP address.

LEAST CONNECTION

- Load balancer checks which server has the less requests.
- Load balancer then sends traffic to the server having the fewest active connections

- Server1 = 100 users
- Server2 = 40 users
- Server3 = 10 users

Next Request goes to → Server3

1. Checks current server load.
2. Sends traffic to less busy server.
3. Best for long user sessions.
4. Improves balancing dynamically.

LEAST RESPONSE TIME

1. Load balancer checks which server has the fast response time.
2. Then it simply sends traffic to the server responding the fastest.

- Server1 → 200ms
- Server2 → 80ms
- Server3 → 50ms

Next Request goes to → Server3

1. Checks server speed continuously.
2. Fast servers get more requests.
3. Improves user experience.
4. Useful for high-performance applications.

RANDOM

Choose a server randomly.

Request 1 → Server 2
Request 2 → Server 1
Request 3 → Server 3
Request 4 → Server 2
Request 5 → Server 3

There is no fixed sequence.

Random selects a server randomly for each request.

WEIGHTED LEAST CONNECTIONS

This combines two ideas:

Weight + Least Connections

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Suppose:
Server 1 → Weight 3
Server 2 → Weight 1

Server 1 is more powerful, so it can handle more connections.

The load balancer considers both:

- Current connections
- Server capacity/weight

Weighted Least Connections considers both server capacity and current connections.

LEAST REQUESTS

This is similar to Least Connections but counts requests currently being handled rather than long-lived connections.

Example:
Server 1 → 20 active requests
Server 2 → 5 active requests
Server 3 → 12 active requests

New request:
→ Server 2

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Least Requests sends traffic to the server handling the fewest active requests.

HEALTH CHECK IN LOAD BALANCER

A health check is a request sent by the Load Balancer to determine whether a backend server is available to receive traffic.

Suppose:
The Load Balancer continuously checks:

1. Server 1 → Health Check → Response → Healthy
2. Server 2 → Health Check → No response → Unhealthy
3. Server 3 → Health Check → Response → Healthy

Therefore:

1. Load balancer only distributes traffic to the healthy instances.

For a web application, it is commonly an HTTP/HTTPS request such as:

1. GET /health
2. GET /healthz
3. GET /api/health

EXAMPLE:

Suppose you have:

1. Server 1 → 10.0.1.10:8080
2. Server 2 → 10.0.2.10:8080
3. Server 3 → 10.0.3.10:8080

Your application provides /Health

When the Load Balancer performs a health check:
LB → Server 1
GET /health
↓
HTTP 200 OK
Server 1 is considered healthy.

WHAT RESPONSE DOES LOAD BALANCER EXPECT

Usually, the Load Balancer checks the HTTP status code.

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EXAMPLE:

1. HTTP 200 - OK (Application is responding correctly)
2. HTTP 500 - Connection timeout
3. HTTP 503 - Connection refused

The exact acceptable status codes depend on the Load Balancer configuration.

LOAD BALANCER DOESN'T CHECK ONCE

The Load Balancer performs health checks repeatedly.

Every 30 seconds -> Health Check -> Server -> 200 OK

Then again:

30 seconds later -> Health Check -> Server -> 200 OK

WHAT HAPPENS WHEN SERVER BECOMES UNHEALTHY

Suppose we have:

1. Server A → Healthy
2. Server B → Healthy
3. Server C → Healthy

Suddenly Server B crashes:

1. Server B → Crash

The Load Balancer detects failed health checks.

1. Server A → Healthy
2. Server B → Unhealthy
3. Server C → Healthy

The Load Balancer stops sending new traffic to Server B.

IMPORTANT HEALTH CHECK

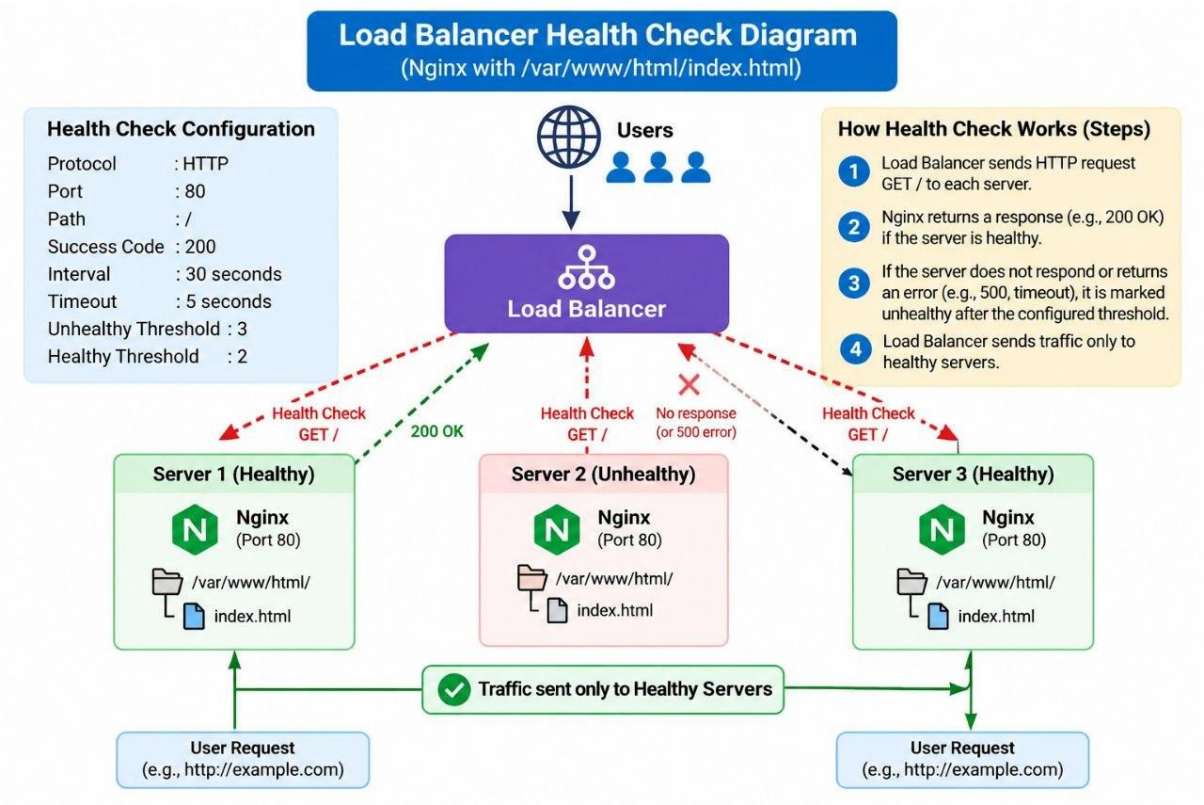
1. Interval -> Every 30 Seconds
   Every 30 seconds load balancer send request to the server

2. Timeout -> 5 seconds
   How long should the Load Balancer wait for a response?
   If the server doesn't respond within 5 seconds, that check can fail.

3. Healthy Threshold -> 2
   Continue two consecutive ok checks declared server healthy

4. Unhealthy Threshold -> 3
   You normally don't want one temporary failure to immediately remove a server.
   Continue Three consecutive failed health checks declared the server unhealthy

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Load Balancer Health Check Diagram
(Nginx with /var/www/html/index.html)
Health Check Configuration
Protocol : HTTP
Port : 80
Path : /
Success Code : 200
Interval : 30 seconds
Timeout : 5 seconds
Unhealthy Threshold : 3
Healthy Threshold : 2
Users
Load Balancer
How Health Check Works (Steps)
1 Load Balancer sends HTTP request GET / to each server.
2 Nginx returns a response (e.g., 200 OK) if the server is healthy.
3 If the server does not respond or returns an error (e.g., 500, timeout), it is marked unhealthy after the configured threshold.
4 Load Balancer sends traffic only to healthy servers.
Health Check GET /
200 OK
Health Check GET /
No response (or 500 error)
Health Check GET /
Server 1 (Healthy)
Nginx (Port 80)
/var/www/html/
index.html
Server 2 (Unhealthy)
Nginx (Port 80)
/var/www/html/
index.html
Server 3 (Healthy)
Nginx (Port 80)
/var/www/html/
index.html
Traffic sent only to Healthy Servers
User Request
(e.g., http://example.com)
User Request
(e.g., http://example.com)

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